

TATA DATA VISUALISATION

1. Project Name

Tata Data Visualisation: Empowering Business with Effective Insights

2. Project Description

This project was completed as part of the Tata Data Visualisation Virtual Experience. The objective was to analyse an online retail dataset and transform the data into clear, business-focused visual insights for senior stakeholders. The analysis focused on revenue trends, countries with growth opportunities, high-value customers, and geographic demand.

3. Technologies and Approach

- **Data:** Online Retail transaction dataset.
- **Data preparation:** Removed records with negative quantities and negative unit prices to support meaningful analysis.
- **Visualisation:** Time-series analysis, country ranking, customer ranking, and geographic/map visualisation.
- **Business approach:** Converted the visual findings into practical recommendations for the CEO and CMO.
- **Key Business Questions Addressed**

Questions Business Requirement

- Q1** How does revenue trend over time, and are there significant seasonal patterns?
- Q2** Which top 10 countries generate the highest revenue, together with quantity sold, excluding the United Kingdom?
- Q3** Who are the top 10 customers by revenue, and how does revenue change from the highest to lower-value customers?
- Q4** Which countries/regions show the greatest product demand and potential opportunities for business expansion, excluding the United Kingdom?

4. Key Analysis and Insights

• Revenue Trend and Seasonality

The first eight months show relatively stable revenue, with average monthly revenue of approximately USD 685K. Revenue begins to increase in September, rising by about 40% compared with the previous month. The upward trend continues into November, when revenue reaches approximately USD 1.5 million, the highest level observed in the available data. December data was incomplete, so no conclusion was drawn for that month. Overall, the pattern suggests that retail sales are strongly influenced by seasonality, particularly toward the final months of the year.

- **Top Countries and Growth Opportunities**

The country analysis identified several markets with strong revenue and sales activity. The United Kingdom was excluded from the growth-opportunity visual because it already has high demand. Countries such as the Netherlands, Ireland, Germany and France showed high volumes of units purchased and revenue generated. These markets represent important opportunities where focused marketing and customer-retention strategies could help increase sales.

- **Top 10 Customers by Revenue**

The customer analysis showed that revenue is distributed across the top 10 customers rather than being concentrated in only one or two customers. The highest-value customer generated about 17% more revenue than the second-highest customer. This indicates relatively low bargaining power among individual customers and a healthy customer-revenue distribution. Maintaining satisfaction among these high-value customers remains important for protecting repeat business.

- **Geographic Demand and Expansion**

The geographic analysis showed that much of the strongest demand is concentrated in European markets, with the Netherlands, Ireland, Germany, France and Australia contributing significant revenue. The analysis also highlighted comparatively lower demand in parts of the American region, Africa, Asia and Russia. These lower-demand regions could be considered for targeted expansion strategies, subject to further market research and validation.

5. Business Recommendations

- Prepare inventory and marketing campaigns ahead of the strong seasonal period observed toward the end of the year.
- Continue strengthening high-performing European markets through targeted campaigns, retention activities and localized offers.
- Maintain strong relationships with the top revenue-generating customers while continuing to diversify the customer base.
- Investigate lower-performing geographic markets before expansion, using localized pricing, product demand and customer research.
- Use the dashboard and visual analysis as a recurring management tool to monitor revenue, customer value and regional performance.

6. Project Outcome

The project demonstrates the ability to translate transactional retail data into understandable visual insights and business recommendations. It addresses the requirements of the Tata virtual experience by combining data preparation, visual analysis, stakeholder-focused questions, and actionable conclusions.

7. GitHub Repository

GitHub Link: https://github.com/Mahanand4/Tata_Data_Visualisation

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